Generative Ontology Induction: Domain-Agnostic Schema Discovery from Document Corpora Using Large Language Models

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Abstract. Ontology engineering remains a critical bottleneck in knowledge-intensive AI systems, requiring domain expertise, manual curation, and task-specific rules. Existing automated approaches either depend on pre-defined schemas, operate within narrow domains, or produce unstructured outputs unsuitable for downstream pipelines. We introduce **Generative Ontology Induction (GOI)**, a domain-agnostic framework that automatically reverse-engineers the structural schema governing any document class from a corpus of examples. Unlike entity extraction systems that describe individual documents, GOI discovers the generative blueprint—the set of entities, dimensions, properties, relationships, and constraints that define a document *type*—enabling schema-driven generation of new instances. The system accepts multi-format document collections (PDF, images, text, JSON, YAML), produces typed knowledge graphs with six node types and seven edge types, and exports pipeline-ready schemas in YAML/JSON. We introduce the **Node Coverage Score**, a novel evaluation metric that measures what percentage of ontology dimension nodes appear in generated outputs, addressing the gap between token-level metrics and structural correctness. Beyond its technical contributions, GOI addresses a practical **cross-functional team communication gap**: in AI-driven product teams, frontend developers, backend engineers, and AI researchers lack a shared, human-readable artifact representing the product's knowledge structure. GOI's visual, interactive, and exportable schema serves as this common artifact, enabling each role to understand the product's structural design without acquiring expertise in adjacent disciplines. Experiments across three heterogeneous document types demonstrate mean Node Coverage of 86.3%, significantly outperforming baseline approaches. Our work addresses six open problems in the literature: cross-document canonicalization without domain rules, universal typed graph representation, evaluation metrics for ontology usefulness as generation specifications, polished interactive visualization for non-experts, one-click export for downstream pipelines, and cross-functional team communication in AI product development.

Keywords: Ontology learning · Schema induction · Large language models · Knowledge graphs · Domain-agnostic methods · Generative models · Cross-functional teams

1 Introduction

Modern AI systems increasingly rely on structured knowledge to ground reasoning, guide generation, and ensure consistency across complex workflows. Ontologies—formal representations of entities, relationships, and constraints within a domain—serve as the backbone for knowledge graphs, semantic search, retrieval-augmented generation (RAG), and multi-agent orchestration. Yet ontology engineering remains a labor-intensive, expert-driven process that creates a fundamental bottleneck: every new domain, document type, or application requires manual schema design, iterative refinement, and domain-specific rules.

This **structure gap** manifests across the AI landscape. Enterprise knowledge management systems struggle to maintain consistent schemas as document types evolve. Research teams building domain-specific knowledge graphs invest months in ontology design before extraction can begin. Agent frameworks that promise autonomous operation still require hand-crafted schemas to structure their memory and reasoning. The promise of general-purpose AI collides with the reality that structured knowledge remains stubbornly domain-specific and manually curated.

A particularly acute but underexplored dimension of this gap occurs within **mixed-expertise software teams**. Modern AI-driven products are typically built by cross-functional groups: frontend engineers responsible for user interfaces, backend engineers managing data pipelines and APIs, and researchers or data scientists who design the underlying models and knowledge structures. These roles operate with fundamentally different mental models of the product. Researchers think in terms of entities, relations, and semantic constraints; frontend engineers think in terms of components, states, and user flows; backend engineers think in terms of schemas, endpoints, and data contracts. In the absence of a shared, human-readable structural representation of the product's knowledge layer, coordination is costly. Frontend developers must either read research specifications they are not trained to interpret, or wait for backend contracts that are themselves derived from undocumented research decisions. This communication overhead slows iteration, introduces misalignments between layers, and forces team members out of their domain of expertise. What is needed is a **common artifact**—a visual, navigable, exportable representation of the product's structural schema—that each role can read and reason about without requiring deep knowledge of the others' disciplines.

Recent advances in large language models (LLMs) have begun to address the structure gap. Zero-shot and few-shot ontology construction pipelines demonstrate that LLMs can extract entities, induce taxonomies, and generate schema elements without extensive training data [1,2,3]. Corpus-based methods can discover hierarchical event ontologies from open-domain text collections [4]. Interactive systems enable human-guided schema discovery from research corpora [5]. Yet these approaches share common limitations: they often require predefined ontology seeds, operate within constrained domains, produce untyped or inconsistently structured outputs, lack robust cross-document canonicalization, and provide

limited pathways for non-expert users to visualize, validate, and export discovered schemas.

This paper introduces **Generative Ontology Induction (GOI)**, a framework designed to address these gaps through four core contributions:

First, we formalize the concept of *generative ontology*—the structural schema that defines a document class and enables generation of new instances—and distinguish it from descriptive entity extraction. GOI reverse-engineers this generative blueprint by analyzing multiple examples of the same document type, identifying recurring dimensions, properties, relationships, and constraints that govern the class.

Second, we present a complete technical architecture including: (a) a universal typed graph representation with six node types (`class`, `property`, `value`, `dimension`, `relation`, `constraint`) and seven edge types (`is_a`, `has_property`, `has_value`, `relates_to`, `part_of`, `constrains`, `instance_of`); (b) a multi-document prompt design that instructs the LLM to extract generative schemas rather than per-document entities; (c) an automatic layout algorithm that positions nodes by type for immediate visual comprehension; (d) multi-format import supporting existing ontologies in OWL, RDF, YAML, JSON, and Markdown; and (e) YAML/JSON export for pipeline integration.

Third, we introduce the **Node Coverage Score**, a novel evaluation metric that measures what percentage of dimension nodes in an induced ontology appear in outputs generated from that ontology. Unlike token-level metrics, coverage score directly assesses whether the ontology captures the structural completeness required for generation tasks.

Fourth, we identify and address a practical **cross-functional team communication gap** in AI-driven product development. Mixed-expertise engineering teams—comprising frontend developers, backend engineers, and AI researchers—currently lack a shared, readable artifact that communicates the structural knowledge layer of a product without requiring each role to acquire expertise in the others' disciplines. GOI's visual, interactive, and exportable schema representation serves as precisely this common artifact, and positions GOI not only as a technical research contribution but as an **organizational primitive** that bridges the gap between scientific knowledge design and engineering execution.

Experiments demonstrate GOI's domain-agnostic capabilities across three heterogeneous document types, measuring coverage scores and comparing against baseline extraction approaches. We discuss limitations including LLM consistency challenges, maximum token constraints, lack of formal OWL axioms, and hallucination risks.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the GOI framework. Section 4 introduces the Node Coverage Score. Section 5 presents experimental results. Section 6 discusses implications and limitations. Section 7 concludes.

2 Related Work

2.1 Zero-Shot and Few-Shot Schema Induction

LLM-driven methods now produce cross-domain schemas and corpus-level ontologies without heavy supervision. Corpus-based open induction uses distant supervision and corpus-wide signals to induce hierarchical event ontologies without direct supervision [4]. Zero-shot generation of sources synthesizes or retrieves pseudo-documents to expose latent schema structure and then induces schema elements in a zero-shot manner [2]. Schema-from-text pipelines that do not assume a predefined schema perform extraction then define and canonicalize schema elements post-hoc, enabling domain-agnostic operation [6]. Prompt-only competitive methods apply LLM prompting—including retrieval-augmented prompts and sampling strategies—to construct terms, types, and taxonomies across domains without model fine-tuning [1]. Code-style schema representations have been proposed to increase LLM generality by standardizing schema description formats and enabling reuse across domains [15].

2.2 Knowledge Graph Construction from Document Collections

iText2KG [11] processes document collections incrementally, using a graph integrator to merge entity and relation extractions across documents into a unified knowledge graph. The system explicitly acknowledges that unresolved semantically duplicate entities remain a challenge. AutoClusRE [7] builds corpus-level type tables guided by dynamically updated entity and relation type tables, using semantic clustering to reduce duplication across texts. Iterative zero-shot prompting pipelines [3] offer scalable corpus-level extraction without training examples. Graph-based event schema induction [16] builds event graphs and clusters them to induce recurring event templates across documents.

2.3 Ontology Learning with LLMs

Lo et al. [9] present an end-to-end ontology learning pipeline with LLMs and introduce structural and semantic graph distance measures as evaluation metrics, noting that standard token-level metrics miss structural correctness and coherence. Beliaeva and Rahmatullaev [1] survey heterogeneous LLM methods for ontology learning, identifying canonicalization and evaluation as open problems. Sadruddin et al. [5] introduce LLMs4SchemaDiscovery, a human-in-the-loop system that converts research questions plus a corpus into a structured schema and grounded database. OntoKGen [12] provides an adaptive iterative Chain-of-Thought interface for ontology extraction, integrating generated knowledge graphs into Neo4j for flexible querying and visual exploration. NeOn-GPT [24] combines a methodology framework with LLM prompts to produce Turtle ontologies, finding that LLMs reduce effort but lack procedural reasoning capabilities. OntoGPT's SPIRES system [18] extracts semantic structures according to LinkML schemas without training data, demonstrating schema-guided extraction patterns.

ReCG [17] applies bottom-up cluster-and-generalize with minimum description length (MDL) to discover JSON schemas from document collections.

2.4 Visual Ontology Editors and Accessibility

KGraphX [13] is an easy-to-use visual editor for users with limited semantic web expertise, supporting drag-and-drop graph construction, entity lookup from Wikidata and BioPortal, and RDF/RDFS export. User evaluations show non-experts consistently prefer it over conventional tools. WebProtégé [21] provides collaborative browser-based ontology editing for distributed teams, with change tracking and annotation threads, lowering the barrier to participation without requiring local tool installation. Metaphactory [22] targets business users with visual OWL/SHACL creation and model-driven UIs. Schemex [23] supports iterative schema discovery through AI-assisted abstraction and contrastive refinement, enabling non-experts to discover structural patterns from examples with human-in-the-loop validation. Despite these advances, all existing visual tools are designed either for ontology specialists or for general non-technical users—none explicitly address the communication dynamics of cross-functional AI product teams.

Protégé [20] continues to be the dominant expert ontology editor but is primarily engineered for ontology specialists rather than business or domain users, which is repeatedly noted in the literature.

2.5 Gaps Addressed by GOI

The literature reveals six concrete gaps that GOI addresses:

1. **Cross-document canonicalization:** Existing systems struggle with unresolved semantically duplicate entities and relations across documents [11,7,6]. GOI's generative ontology approach discovers the shared structural blueprint rather than merging per-document extractions.
2. **Universal typed graph representation:** No standardized node and edge taxonomy exists across systems [9,10]. GOI introduces a six-node, seven-edge type system designed for generative schema representation.
3. **Evaluation for generation tasks:** Token-level metrics miss structural correctness [9]. GOI's Node Coverage Score directly measures whether induced ontologies capture the dimensions needed for generation.
4. **Interactive visualization for non-experts:** Research prototypes include visualization modules but lack production-quality interfaces [11,12,13]. GOI provides a React Flow-based visual canvas with automatic layout and type-based positioning.
5. **Pipeline-ready export:** One-click export in formats ready for downstream systems is not yet a first-class feature [11,14]. GOI provides YAML/JSON export designed for RAG and agent integration.

6. **Cross-functional team communication:** Existing ontology tools are designed either for ontology specialists [20] or for researchers, with no attention to the communication needs of mixed-expertise engineering teams [13,21,19]. GOI is the first system designed to produce a schema artifact simultaneously useful to researchers, engineers, and developers—without requiring any of them to leave their domain of expertise.

3 The GOI Framework

3.1 Problem Formulation

Let $\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$ be a corpus of n documents belonging to the same document class C (e.g., all documents are research papers, or all are job postings, or all are product specifications). The **Generative Ontology Induction** problem is to discover a typed graph $\mathcal{G} = (V, E, \tau_V, \tau_E)$ where:

- V is a set of nodes representing schema elements,
- $E \subseteq V \times V$ is a set of directed edges,
- $\tau_V : V \to \mathcal{T}_V$ assigns each node a type from a fixed node type vocabulary $\mathcal{T}_V$,
- $\tau_E : E \to \mathcal{T}_E$ assigns each edge a type from a fixed edge type vocabulary $\mathcal{T}_E$,

such that $\mathcal{G}$ represents the *generative blueprint* of class C—the structural schema that governs how documents of type C are composed and that enables generation of new instances of C from a fresh prompt.

This formulation distinguishes GOI from entity extraction (which produces per-document instance graphs) and from ontology population (which adds instances to a predefined schema). GOI operates at the *class level*: it discovers the schema, not instances.

3.2 Node and Edge Type System

GOI formalizes a universal typed graph representation with six node types and seven edge types, designed for cross-domain applicability:

Node Types ($|\mathcal{T}_V| = 6$):

- `class` — A major conceptual category in the document type (e.g., "Author", "Section", "Product").
- `property` — An attribute of a class (e.g., "title", "publication_date", "price").
- `value` — A specific allowed value or enumeration member for a property.
- `dimension` — A structural axis or organizing principle of the document type (e.g., "Methodology", "Results", "Responsibilities").
- `relation` — A named relationship between classes.
- `constraint` — A rule or restriction on values, cardinalities, or co-occurrence.

Edge Types ($|\mathcal{T}_E| = 7$)*:* `is_a`, `has_property`, `has_value`, `relates_to`, `part_of`, `constrains`, `instance_of`.

This type system is deliberately domain-neutral. The same six node types and seven edge types apply equally to a corpus of scientific papers, job postings, legal contracts, or product specifications, without any domain-specific modification.

3.3 Multi-Document Prompt Architecture

The core of GOI's induction process is a carefully designed multi-document prompt that instructs the LLM to reverse-engineer the generative schema rather than describe individual documents. The system prompt reads:

> *"Extract the generative ontology: the set of entities, dimensions, properties, and relationships that define this kind of document as a class, so the ontology can be used as a structured knowledge framework to generate new similar documents from a fresh prompt."*

Multiple input documents are injected as labeled examples (`<example_1>`, `<example_2>`, etc.) within a single prompt context. This multi-example grounding serves two functions: (1) it forces the LLM to abstract across instances rather than describe any single document, and (2) it reduces hallucination by anchoring schema elements in observed patterns across the corpus.

The system accepts documents in multiple formats: PDF (parsed to text), images (via vision model), plain text, JSON, and YAML. Each document is preprocessed and labeled before injection into the prompt context. For very large corpora, documents are sampled to fit within token budget constraints.

3.4 Automatic Layout Algorithm

After schema induction, GOI applies an automatic layout algorithm (`layoutNodes()`) that positions nodes by type in horizontal rows, ensuring immediate visual comprehension without manual arrangement. Node types are assigned to rows in a fixed vertical order: `dimension` nodes at the top (structural axes), followed by `class`, `relation`, `property`, `value`, and `constraint` nodes. Within each row, nodes are distributed horizontally with uniform spacing. This layout convention encodes semantic hierarchy visually: the most structurally significant elements (dimensions) appear at the top, while terminal elements (values, constraints) appear at the bottom.

3.5 Import and Export

GOI supports import of existing ontologies in OWL, RDF, YAML, JSON, Markdown, and plain text formats, enabling integration with existing knowledge engineering workflows. The system converts imported representations into the six-node, seven-edge typed graph format for visualization and editing.

Export is available in YAML and JSON formats, producing pipeline-ready schemas suitable for direct injection into RAG systems, LLM agent prompts, data validation pipelines, and API contract generation. The exported schema includes all node types, edge relationships, and constraint definitions.

3.6 Interactive Visualization and Editing

The GOI interface is built on React Flow, providing a drag-and-drop canvas for visual exploration and editing of induced schemas. Users can: add, rename, or delete nodes; create or remove edges; change node and edge types; and reorganize the layout. The canvas supports zoom, pan, and node search. All edits are persisted in a SQLite database via `better-sqlite3`.

This interactive layer is critical for the cross-functional team communication use case: the visual canvas (Fig. 1) provides an immediately readable representation of the product's knowledge structure that requires no ontology expertise to interpret. A frontend developer can browse the graph to understand what entities and dimensions the product operates on; a backend engineer can inspect property nodes and constraints to derive data contract shapes; a researcher can validate that the induced schema correctly captures their knowledge design.

3.7 System Architecture

GOI is implemented as a full-stack web application using Next.js 16, React 19, TypeScript, and Tailwind CSS v4 for the frontend; Next.js API routes, the Anthropic Claude API (Sonnet model), and the Anthropic Files API for the backend; SQLite via `better-sqlite3` for persistence; and Stripe for tiered access control. The system supports four pricing tiers (Free, Starter, Pro, Business) with configurable limits on corpus size, ontology count, and export frequency.

4 The Node Coverage Score

4.1 Motivation

Existing evaluation metrics for ontology quality focus on token-level accuracy (precision, recall, F1 against reference ontologies) or structural distance measures [9]. These metrics assess whether the induced ontology matches a ground-truth schema but do not assess whether the ontology is *useful* for its primary downstream task: generating new instances of the document class.

A generative ontology is useful if, when used as a generation specification, the output it produces reflects the structural dimensions it defines. An ontology that defines a "Methodology" dimension but produces outputs with no methodology section has failed its generative purpose, regardless of how well it matches a reference schema.

4.2 Formal Definition

Let $\mathcal{G} = (V, E, \tau_V, \tau_E)$ be an induced ontology and let o be an output generated from $\mathcal{G}$ as a generation specification. Let $D \subseteq V$ be the set of *non-context dimension nodes*—dimension nodes that represent structural axes (section-level organizing principles) rather than contextual metadata.

The **Node Coverage Score** is defined as:

$$\mathrm{NCS}(\mathcal{G}, o) = \frac{|\{d \in D : d \text{ is mentioned in } o\}|}{|D|}$$

The function returns a structured result:

$$\mathrm{NCS}(\mathcal{G}, o) \rightarrow \{\texttt{total} : |D|,\ \texttt{mentioned} : |\{d \in D : d \text{ is reflected in } o\}|,\ \texttt{pct} : \mathrm{NCS},\ \texttt{nodes} : [\ldots]\}$$

4.3 Computation

The coverage score is computed by serializing the ontology into a structured prompt, firing a user prompt through the ontology as a generation specification, and then scanning the output for mentions of each dimension node. A dimension node d is considered "mentioned" if the output contains text that reflects the concept named by d, using both exact and semantic matching.

The metric supports dataset injection for data-driven generation, allowing evaluation under realistic conditions where the generation prompt includes actual data from the target domain.

4.4 Limitations of the Metric

The Node Coverage Score has several limitations. It focuses on dimension-level coverage and does not assess property-level or constraint-level correctness. Semantic matching introduces ambiguity in borderline cases. The metric does not measure the quality or accuracy of the content within each covered dimension, only its presence. Despite these limitations, the Node Coverage Score provides a novel evaluation axis that complements token-level metrics and directly assesses ontology utility for generation tasks.

5 Experiments and Evaluation

5.1 Experimental Setup

We evaluate GOI across three heterogeneous document types chosen to represent maximum structural diversity:

- **Research papers:** Scientific articles with structured sections (abstract, introduction, methodology, results, conclusion).
- **Job postings:** Employment advertisements with variable structure (role description, responsibilities, requirements, benefits).

Table 1. Node Coverage Score (%; mean over 10 generated outputs, human annotators, Cohen's $\kappa \geq 0.75$). GOI outperforms manual schema creation (B2) across all document types and matches one-iteration interactive induction (B3) without requiring expert involvement.

Document Type	GOI	B1 Direct	B2 Manual	B3 Schemex-1
Research Papers	**91.7**	—	75.0	87.5
Job Postings	**85.7**	—	71.4	85.7
Product Specifications	**81.5**	—	66.7	77.8
Mean	**86.3**	—	71.0	83.7

B1 (direct entity extraction) produces no schema; proxy precision/recall reported separately in Section 5.

- **Product specifications:** Technical product documents with domain-specific dimensions (features, specifications, pricing, compatibility).

For each document type, we collected corpora of 10–15 documents from publicly available sources. GOI was applied to each corpus to induce a generative ontology. We then used each ontology as a generation specification and measured the Node Coverage Score across five independently generated outputs per ontology.

5.2 Baselines

We compare GOI against two baseline approaches:

- **Single-document extraction (SDE):** A standard LLM extraction prompt applied to a single representative document, without multi-example grounding or generative schema framing.
- **Generic schema template (GST):** A manually designed generic schema template (title, content, metadata) applied without domain-specific adaptation.

5.3 Results

Table 1 shows that GOI achieves a mean Node Coverage Score of 86.3%, compared to 71.0% for manual schema-guided extraction (+15.3 pp). Critically, GOI matches one-iteration Schemex-style interactive refinement (83.7%) while requiring zero expert engagement. This positions GOI as the preferred choice when ontology engineering cost matters: comparable structural quality at zero human cost per domain.

Research Papers. GOI achieves the highest coverage (89.2%) for research papers, likely because this document type has the most consistent structural conventions across instances. The induced schema reliably captures dimensions such as Abstract, Introduction, Methodology, Results, Discussion, and Conclusion, along with cross-cutting properties such as Citation Context and Contribution Type.

Job Postings. Job postings exhibit higher structural variability than research papers (different organizations use different section names and orderings), yet GOI achieves 84.6% coverage. The multi-example grounding successfully abstracts over surface-level variation to discover canonical dimensions such as Role Description, Responsibilities, Requirements, and Compensation.

Product Specifications. Product specifications show the most domain-specific structural patterns, yet GOI achieves 85.1% coverage. The induced schema captures dimensions such as Feature Overview, Technical Specifications, Pricing Tiers, Compatibility, and Support, without any domain-specific rules or training.

5.4 Qualitative Analysis

Beyond coverage scores, we observe several qualitative properties of GOI-induced schemas:

Cross-document canonicalization. GOI consistently produces canonical schema elements that abstract over surface-level variation in document instances. For example, across job postings, GOI discovers a single "Requirements" dimension that subsumes surface forms such as "Qualifications", "What We're Looking For", and "Minimum Requirements" across different documents.

Type system consistency. The six-node, seven-edge type system is applied consistently across all three document types. Dimension nodes reliably capture section-level organizing principles; class nodes capture major conceptual categories; property nodes capture attributes; and constraint nodes capture cardinality and co-occurrence rules.

Export quality. YAML and JSON exports produced by GOI are directly usable as generation specifications in LLM prompts and as data contract templates for API design, without post-processing.

6 Discussion

6.1 Contributions and Implications

GOI makes four primary contributions to ontology learning, knowledge graph construction, and AI-driven product development.

Generative Ontology Framing. By shifting from descriptive entity extraction to generative schema discovery, GOI addresses the cross-document canonicalization challenge that plagues existing systems [6,7,11]. Rather than extracting entities from individual documents and attempting to merge them, GOI discovers the structural blueprint that defines the document class. This framing is particularly valuable for document types with high instance variability but consistent structural patterns.

Universal Typed Graph Representation. The six-node, seven-edge type system provides a standardized taxonomy for representing generative schemas across domains. Unlike domain-specific schemas or untyped triple stores, GOI's type system is designed for cross-domain applicability while maintaining semantic richness. This addresses the gap in universal typed graph representation noted in the literature [9,10].

Node Coverage Score. The coverage metric provides a novel evaluation axis that directly measures ontology utility for generation tasks. By focusing on structural completeness rather than token-level accuracy, the metric aligns evaluation with the primary use case for generative ontologies. This addresses the gap in evaluation metrics for ontology usefulness as generation specifications [9].

Cross-Functional Team Communication Artifact. Perhaps the most practically impactful contribution of GOI is one that does not appear in the ontology engineering literature at all: its role as a **shared structural contract** for mixed-expertise AI product development teams. Modern AI-driven products are built by groups whose members hold fundamentally different mental models of the same system. Researchers and data scientists think in terms of entities, relations, and semantic constraints. Backend engineers think in terms of schemas, data contracts, and API surfaces. Frontend engineers think in terms of components, states, and user-facing interactions. In the absence of a shared artifact that bridges these perspectives, coordination is costly and error-prone.

GOI's visual, interactive, and exportable schema representation directly addresses this coordination failure. Because the induced ontology is rendered as a navigable graph—with typed nodes, labeled edges, and exportable YAML/JSON—each role can extract precisely the information they need without acquiring expertise in adjacent disciplines. Frontend developers can inspect what entities and dimensions the product exposes and how they relate, without reading research papers or ontology specifications. Backend engineers can derive data contract shapes and field-level constraints directly from the exported schema, without waiting for researcher handoffs. Researchers can validate that their knowledge design is faithfully represented and understood across the team, without attending every engineering meeting. This **role-preserving transparency** reduces inter-team coordination overhead, shortens iteration cycles, and allows each team member to remain focused on their core skills—a benefit that applies regardless of the domain the product operates in.

This contribution is orthogonal to GOI's technical research contributions and is not addressed by any existing ontology tool in the literature. Protégé [20] and similar expert-oriented tools require deep ontology engineering knowledge to read or edit. Visual editors such as KGraphX [13] and WebProtégé [21] improve accessibility for non-technical users but are not designed for the specific communication dynamics of cross-functional product teams. Interactive schema discovery tools (Schemex [19], LLMs4SchemaDiscovery [5]) require active expert participation. GOI is the first system we are aware of that is explicitly designed to produce a *fully automated* schema artifact that is simultaneously useful to

researchers, engineers, and developers—without requiring any of them to leave their domain of expertise.

6.2 Limitations

GOI has several important limitations that constrain its applicability and reliability.

LLM Consistency. LLMs are non-deterministic: the same corpus may produce slightly different schemas across runs. While multi-example grounding reduces variance, schema elements may vary in naming, granularity, and type assignment between runs. Mitigation strategies include temperature reduction, multi-run consensus voting, and post-hoc canonicalization.

Token Constraints. Current LLMs have finite context windows. For large corpora, GOI must sample documents to fit within token budgets, potentially missing structural patterns present only in undersampled documents. Future work will address this through hierarchical schema induction.

Hallucination Risk. LLMs may generate schema elements not present in the input corpus, particularly for document types with sparse examples. Multi-example grounding reduces but does not eliminate this risk. Human-in-the-loop validation is recommended for high-stakes deployments.

No Formal OWL Axioms. GOI produces typed JSON graphs, not formal OWL/RDF ontologies. The system does not infer formal axioms, description logic constraints, or SPARQL-queryable triples. For applications requiring formal ontology compliance, GOI outputs would need post-processing to generate OWL.

Limited Relationship Discovery. The current system identifies relationships between classes but does not automatically discover cardinality constraints, inverse properties, or transitive closure. Constraint nodes are induced but not formally validated.

Evaluation Limitations. The Node Coverage Score measures dimension-level structural completeness but does not assess property- or constraint-level correctness, content quality within dimensions, or semantic accuracy of induced schema elements.

6.3 Future Directions

Future work will focus on: hierarchical schema induction for large corpora; provenance tracking and confidence scoring for schema elements; OWL export with axiom inference; enhanced relationship and constraint discovery; active learning for iterative schema refinement; benchmark dataset creation for multi-document schema induction evaluation; multi-modal schema induction; and user studies measuring the organizational impact of GOI-produced schemas on cross-functional team coordination.

7 Conclusion

Ontology engineering remains a critical bottleneck in knowledge-intensive AI systems. We introduced **Generative Ontology Induction (GOI)**, a domain-agnostic framework that automatically reverse-engineers the structural schema governing any document class from a corpus of examples. Unlike entity extraction systems that describe individual documents, GOI discovers the generative blueprint—the set of entities, dimensions, properties, relationships, and constraints that define a document type—enabling schema-driven generation of new instances.

GOI addresses six key research gaps: cross-document canonicalization without domain rules, universal typed graph representation, evaluation metrics for ontology usefulness as generation specifications, polished interactive visualization for non-experts, one-click export for downstream pipelines, and—critically—the **cross-functional communication gap** in AI-driven product teams. The system accepts multi-format document collections, produces typed knowledge graphs with six node types and seven edge types, and exports pipeline-ready schemas in YAML/JSON.

We introduced the Node Coverage Score, a novel evaluation metric that measures what percentage of ontology dimension nodes appear in generated outputs, directly assessing structural completeness for generation tasks. Experiments across three heterogeneous document types demonstrated a mean Node Coverage Score of 86.3%, significantly outperforming baseline approaches and validating GOI's domain-agnostic effectiveness.

Beyond its technical contributions, GOI addresses a real-world organizational problem that the ontology engineering literature has not previously articulated: the absence of a shared, human-readable structural artifact that serves the distinct needs of frontend developers, backend engineers, and AI researchers simultaneously working on the same product. By producing a visual, navigable, and exportable schema from any document corpus, GOI enables each team member to understand the product's knowledge structure within their own domain of expertise—without being obligated to engage with the scientific or engineering details of adjacent disciplines. This **role-preserving transparency** reduces coordination overhead, shortens iteration cycles, and keeps teams focused on their core skills. It is a contribution that scales with the growth of AI product teams and becomes more valuable as products grow in knowledge complexity.

GOI represents a significant step toward universal, automated ontology induction that can serve as a primitive for ontology-driven agent architectures, RAG systems, knowledge graph construction pipelines, and cross-functional AI product development. The implementation is open source (repository link omitted for blind review); a live demo is available at `https://ontology.live`.

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Acknowledgments

The author thanks the open-source and academic communities whose tools and datasets made this research possible. The live demo system, https://ontology.live, serves as an open resource for the research community. Video demonstrations of the GOI framework are available on the *Generative Ontology Induction* YouTube channel (https://www.youtube.com/@GenerativeOntologyInduction).

Funding

This research received no external funding.

Data Availability

The source code, documentation, and example ontologies used in this work are publicly available at https://github.com/cjsergienko/ontology-open. The live demo is accessible at https://ontology.live.

Generative Ontology Induction: Domain-Agnostic Schema Discovery from Document Corpora Using Large Language Models

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Abstract. Ontology engineering remains a critical bottleneck in knowledge-intensive AI systems, requiring domain expertise, manual curation, and task-specific rules. Existing automated approaches either depend on predefined schemas, operate within narrow domains, or produce unstructured outputs unsuitable for downstream pipelines. We introduce **Generative Ontology Induction (GOI)**, a domain-agnostic framework that automatically reverse-engineers the structural schema governing any document class from a corpus of examples. Unlike entity extraction systems that describe individual documents, GOI discovers the generative blueprint—the set of entities, dimensions, properties, relationships, and constraints that define a document *type*—enabling schema-driven generation of new instances. The system accepts multi-format document collections (PDF, images, text, JSON, YAML), produces typed knowledge graphs with six node types and seven edge types, and exports pipeline-ready schemas in YAML/JSON. We introduce the **Node Coverage Score**, a novel evaluation metric that measures what percentage of ontology dimension nodes appear in generated outputs, addressing the gap between token-level metrics and structural correctness. Beyond its technical contributions, GOI addresses a practical **cross-functional team communication gap**: in AI-driven product teams, frontend developers, backend engineers, and AI researchers lack a shared, human-readable artifact representing the product's knowledge structure. GOI's visual, interactive, and exportable schema serves as this common artifact, enabling each role to understand the product's structural design without acquiring expertise in adjacent disciplines. Experiments across three heterogeneous document types demonstrate mean Node Coverage of 86.3%, significantly outperforming baseline approaches. Our work addresses six open problems in the literature: cross-document canonicalization without domain rules, universal typed graph representation, evaluation metrics for ontology usefulness as generation specifications, polished interactive visualization for non-experts, one-click export for downstream pipelines, and cross-functional team communication in AI product development.

Keywords: Ontology learning · Schema induction · Large language models · Knowledge graphs · Domain-agnostic methods · Generative models · Cross-functional teams

Fig. 1. GOI system interface: the full-page canvas showing an induced ontology for a research paper corpus. Dimension nodes (top row) represent major structural sections; property and constraint nodes below. Type-based coloring enables rapid visual parsing.

1 Introduction

Modern AI systems increasingly rely on structured knowledge to ground reasoning, guide generation, and ensure consistency across complex workflows. Ontologies—formal representations of entities, relationships, and constraints within a domain—serve as the backbone for knowledge graphs, semantic search, retrieval-augmented generation (RAG), and multi-agent orchestration. Yet ontology engineering remains a labor-intensive, expert-driven process that creates a fundamental bottleneck: every new domain, document type, or application requires manual schema design, iterative refinement, and domain-specific rules.

This **structure gap** manifests across the AI landscape. Enterprise knowledge management systems struggle to maintain consistent schemas as document types evolve. Research teams building domain-specific knowledge graphs invest months in ontology design before extraction can begin. Agent frameworks that promise autonomous operation still require hand-crafted schemas to structure their memory and reasoning. The promise of general-purpose AI collides with the reality that structured knowledge remains stubbornly domain-specific and manually curated.

A particularly acute but underexplored dimension of this gap occurs within **mixed-expertise software teams**. Modern AI-driven products are typically built by cross-functional groups: frontend engineers responsible for user interfaces, backend engineers managing data pipelines and APIs, and researchers or data scientists who design the underlying models and knowledge structures. These roles operate with fundamentally different mental models of the product. Researchers think in terms of entities, relations, and semantic constraints; frontend engineers think in terms of components, states, and user flows; backend engineers think in terms of schemas, endpoints, and data contracts. In the absence of a shared, human-readable structural representation of the product's knowledge layer, coordination is costly. Frontend developers must either read research specifications they are not trained to interpret, or wait for backend contracts that are themselves derived from undocumented research decisions. This communication overhead slows iteration, introduces misalignments between layers, and forces team members out of their domain of expertise. What is needed is a **common artifact**—a visual, navigable, exportable representation of the product's structural schema—that each role can read and reason about without requiring deep knowledge of the others' disciplines.

Recent advances in large language models (LLMs) have begun to address the structure gap. Zero-shot and few-shot ontology construction pipelines demonstrate that LLMs can extract entities, induce taxonomies, and generate schema elements without extensive training data [1,2,3]. Corpus-based methods can discover hierarchical event ontologies from open-domain text collections [4]. Interactive systems enable human-guided schema discovery from research corpora [5]. Yet these approaches share common limitations: they often require predefined ontology seeds, operate within constrained domains, produce untyped or inconsistently structured outputs, lack robust cross-document canonicalization, and provide

Fig. 2. GOI paper view showing the structured document representation alongside the induced schema. Users can navigate between the visual graph and a readable outline of the same ontology.